

Artificial Intelligence

Lecture 06 – First Order Logic

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First Order Logic – Representation of Language

- Whorf (1956) suggest that communities determine language categories
- Wanner (1975) subject remember the content of what they read better than actual worlds
- Mitchell et al (2008) could predict – with above chance accuracy – the areas of the brain that would activate with the certain words (fMRI)

First Order Logic – Formal Languages

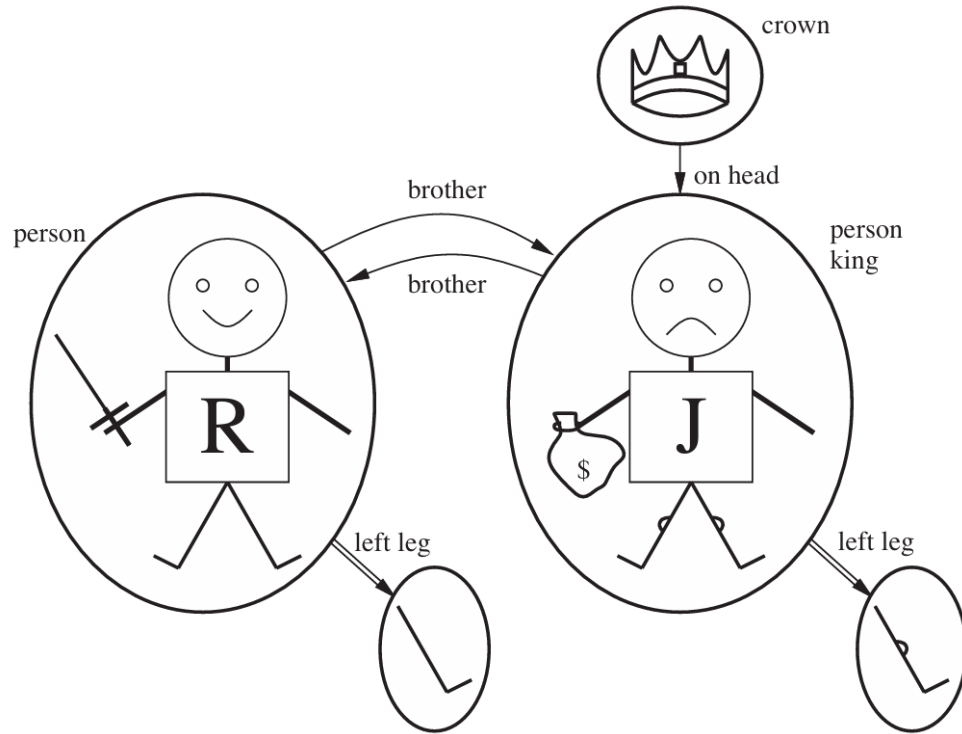
- Object (cat, dog, house, etc)
- Relations (has color, larger than, comes between etc)
- Facts: (one value for a given input: has feathers, can swim)
- Facts have a truth value. True or False

Formal Languages – Ontological and Epistemological Commitments

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

Figure 8.1 Formal languages and their ontological and epistemological commitments.

Relationships – Models for FoL



First Order Logic – Syntax

$$\begin{aligned}
 \textit{Sentence} &\rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence} \\
 \textit{AtomicSentence} &\rightarrow \textit{Predicate} \mid \textit{Predicate}(\textit{Term}, \dots) \mid \textit{Term} = \textit{Term} \\
 \textit{ComplexSentence} &\rightarrow (\textit{Sentence}) \mid [\textit{Sentence}] \\
 &\mid \neg \textit{Sentence} \\
 &\mid \textit{Sentence} \wedge \textit{Sentence} \\
 &\mid \textit{Sentence} \vee \textit{Sentence} \\
 &\mid \textit{Sentence} \Rightarrow \textit{Sentence} \\
 &\mid \textit{Sentence} \Leftrightarrow \textit{Sentence} \\
 &\mid \textit{Quantifier Variable}, \dots \textit{Sentence}
 \end{aligned}$$

$$\begin{aligned}
 \textit{Term} &\rightarrow \textit{Function}(\textit{Term}, \dots) \\
 &\mid \textit{Constant} \\
 &\mid \textit{Variable}
 \end{aligned}$$

$$\begin{aligned}
 \textit{Quantifier} &\rightarrow \forall \mid \exists \\
 \textit{Constant} &\rightarrow A \mid X_1 \mid \textit{John} \mid \dots \\
 \textit{Variable} &\rightarrow a \mid x \mid s \mid \dots \\
 \textit{Predicate} &\rightarrow \textit{True} \mid \textit{False} \mid \textit{After} \mid \textit{Loves} \mid \textit{Raining} \mid \dots \\
 \textit{Function} &\rightarrow \textit{Mother} \mid \textit{LeftLeg} \mid \dots
 \end{aligned}$$

OPERATOR PRECEDENCE : $\neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow$

First Order Logic – More on syntax

- Three kinds of symbols
 - Constant: objects
 - Predicate: relations
 - Function: can return values other than truth vals.
- Predicate and Function have “arity”.
- Symbols have an interpretation
- Terms: *LeftLeg*(John)
- Atomic Sentences state facts: *Brother*(Richard, John)
- Complex Sentence:
 - $\text{Brother}(R, J) \wedge \text{Brother}(J, R)$
 - $\neg \text{King}(\text{Richard}) \rightarrow \text{King}(\text{John})$
- Universal Quantifiers: $\forall \text{King}(x) \rightarrow \text{Person}(x)$
- Existential Quantifiers: $\exists \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$

First Order Logic – examples

What is the interpretation for:

- $\text{King}(\text{Richard}) \vee \text{King}(\text{John})$
- $\neg \text{Brother}(\text{LeftLeg}(\text{Richard}), \text{John})$
- $\forall x \forall y \text{Brother}(x, y) \rightarrow \text{Sibling}(x, y)$
- $\text{In}(\text{Paris, France}) \wedge \text{In}(\text{Marseilles, France})$
- $\forall c \text{Country}(c) \wedge \text{Border}(c, \text{Ecuador}) \rightarrow \text{In}(c, \text{SouthAmerica})$
- $\exists \text{Country}(c) \wedge \text{Border}(c, \text{Spain}) \wedge \text{Border}(c, \text{Italy})$

First Order Logic – examples

Richard has only two brothers, John and Geoffrey:

$\text{Brother}(\text{John}, \text{Richard}) \wedge \text{Brother}(\text{Geoffrey}, \text{Richard}) \wedge \text{John} \neq \text{Geoffrey} \wedge$
 $\forall x \text{Brother}(x, \text{Richard}) \rightarrow (x = \text{John} \vee x = \text{Geoffrey})$

No Region in South America borders any region in Europe

$\forall c, d \text{In}(c, \text{SouthAmerica}) \wedge \text{In}(d, \text{Europe}) \rightarrow \neg \text{Border}(c, d)$

No two adjacent countries have the same map color

$\forall x, y \text{Country}(x) \wedge \text{Country}(y) \wedge \text{Border}(x, y) \rightarrow$
 $\neg (\text{Color}(x) = \text{Color}(y)) \wedge \neg (x = y)$

Assertions and Queries in FoL – TELL and ASK

- TELL(KB, King(John))
- TELL(KB, $\forall x \text{ King}(x) \rightarrow \text{Person}(x)$)
- ASK(KB, King(John)) return True
- ASK(KB, $\exists x \text{ person}(x)$) return True
- ASKVARS(KB, Person(x) yields {x/John, x/Richard}, a binding list

First Order Logic - Kinship

- “The son of my father is my brother”
- “One’s grandmother is the mother of one’s parent”
- Domain: People
- Unary predicates: Male, Female
- Relations:
 - Parent, Sibling, Brother, Sister, Child, Daughter, Son, Spouse, Wife, Husband, Grandparent, Grandchild, Cousin, Aunt, Uncle
- Functions: Mother, Father

First Order Logic - Kinship

- “One’s mother is one’s female parent”

$$\forall m, c \text{ Mother}(c) = m \leftrightarrow \text{Female}(m) \wedge \text{Parent}(m, c)$$

- “A sibling is another child of one’s parent”

$$\forall x, y \text{ Sibling}(x, y) \leftrightarrow x \neq y \wedge \exists p \text{ Parent}(p, x) \wedge \text{Parent}(p, y)$$

- “Wendy is female”
- Female (wendy)
- These are axioms

First Order Logic – Wumpus – Time domain included

- Can be represented more concisely
- At time step 3
Percept([Stench, Breeze, Glitter, None, None], 3)
- At time step 6
Percept([Stench, Breeze, Glitter, None, None], 3)
- Actions can be:
Turn(Right), Turn(Left), Forward, Shoot, Grab, Climb
- And we can ASK the best action at time step 5
ASKVARS($\exists a$ BestAction(a,5))

FOL: Wumpus – Encoding complex rules

- Can encode:
 - Raw percepts:
 $\forall t, s, g, m, c \text{ Percept}([s, b, \text{Glitter}, m, c], t) \rightarrow \text{Glitter}(t)$
 - Reflex actions: $\forall t \text{ Glitter}(t) \rightarrow \text{BestAction}(\text{Grab}, t)$
- Instead of encoding stuff like:
 - $\text{Adjacent}(\text{Square}[1,2], \text{Square}[1,1])$
 - $\text{Adjacent}(\text{Square}[3,4], \text{Square}[4,4])$
- Encode:
 $\forall x, y, a, b, \text{Adjacent}([x, y], [a, b]) \leftrightarrow (x=a \wedge (y=b-1 \vee y=b+1)) \vee (y=b \wedge (x=a-1 \vee x=a+1))$

Creating Knowledge Base in FoL

- Identify the Task
- Assemble the relevant knowledge
- Decide on a vocabulary of predicates, functions and constants
- Encode general knowledge about the domain (rules)
- Encode a description of the problem
- Pose queries to the inference procedure and get answers
- Debug the KB

Inference in FoL

- Given $\forall x \text{ King}(x) \wedge \text{Greedy}(x) \rightarrow \text{Evil}(x)$
- One can infer
- $\text{King}(\text{John}) \wedge \text{Greedy}(\text{John}) \rightarrow \text{Evil}(\text{John})$
- $\text{King}(\text{Richard}) \wedge \text{Greedy}(\text{Richard}) \rightarrow \text{Evil}(\text{Richard})$
- $\text{King}(\text{Father}(\text{John})) \wedge \text{Greedy}(\text{Father}(\text{John})) \rightarrow \text{Evil}(\text{Father}(\text{John}))$

Inference in FoL

- Universal Instantiation (in a $\forall$ rule, substitute all symbols)
- Existential Instantiation (in a $\exists$ rule, substitute one symbol, use the rule and discard)
- Skolem Constants is a new variable that represents a new inference.

Inference in FoL

- Suppose KB:
 - $\forall x \text{ King}(x) \wedge \text{Greedy}(x) \rightarrow \text{Evil}(x)$
 - $\text{King}(\text{John})$
 - $\text{Greedy}(\text{John})$
 - $\text{Brother}(\text{Richard}, \text{John})$
- Apply UI using $\{x/\text{John}\}$ and $\{x/\text{Richard}\}$
 - $\text{King}(\text{John}) \wedge \text{Greedy}(\text{John}) \rightarrow \text{Evil}(\text{John})$
 - $\text{King}(\text{Richard}) \wedge \text{Greedy}(\text{Richard}) \rightarrow \text{Evil}(\text{Richard})$
- And discard the Universally quantified sentence. We can get the KB to be propositions.

Inference in FoL

- Suppose KB:
 - $\forall x \text{ King}(x) \wedge \text{Greedy}(x) \rightarrow \text{Evil}(x)$
 - $\text{King}(\text{John})$
 - $\forall y \text{ Greedy}(y)$
- Apply UI using $\{x/\text{John}\}$ and $\{y/\text{John}\}$

Inference – Generalized Modus Ponens

For atomic sentences p_i , p_i' , and q , where there is a substitution θ such that $\text{SUBST}(\theta, p_i') = \text{SUBST}(\theta, p_i)$, for all i ,

$$\frac{p_1', p_2', \dots, p_n', (p_1 \wedge p_2 \wedge \dots \wedge p_n \Rightarrow q)}{\text{SUBST}(\theta, q)} .$$

There are $n + 1$ premises to this rule: the n atomic sentences p_i' and the one implication. The conclusion is the result of applying the substitution θ to the consequent q . For our example:

p_1' is <i>King(John)</i>	p_1 is <i>King(x)</i>
p_2' is <i>Greedy(y)</i>	p_2 is <i>Greedy(x)</i>
θ is $\{x/\text{John}, y/\text{John}\}$	q is <i>Evil(x)</i>
$\text{SUBST}(\theta, q)$ is <i>Evil(John)</i> .	

Inference - Unification

$\text{UNIFY}(p, q) = \theta$ where $\text{SUBST}(\theta, p) = \text{SUBST}(\theta, q)$

For example:

- We ask $\text{ASKVARS}(\text{Knows}(\text{John}, x))$ (Whom does John know?)
- $\text{UNIFY}(\text{Knows}(\text{John}, x), \text{Knows}(\text{John}, \text{Jane})) = \{x/\text{Jane}\}$
- $\text{UNIFY}(\text{Knows}(\text{John}, x), \text{Knows}(y, \text{Bill})) = \{y/\text{John}, x/\text{Bill}\}$
- $\text{UNIFY}(\text{Knows}(\text{John}, x), \text{Knows}(y, \text{Mother}(y))) = \{y/\text{John}, x/\text{Mother}(\text{John})\}$

Algorithm in the book (goes variable by variable recursively unifying)

Inference – Putting it all together

"The Law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, and enemy of America, has some missiles , and all of its missiles were sold to it by Colonel West, who is American"

Prove that Colonel Wes is a Criminal

Inference – The KB

The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.

$American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z) \Rightarrow Criminal(x)$

$Owns(Nono, M_1)$

$Missile(M_1)$

$Missile(x) \wedge Owns(Nono, x) \Rightarrow Sells(West, x, Nono)$

$Missile(x) \Rightarrow Weapon(x)$

$Enemy(x, America) \Rightarrow Hostile(x)$

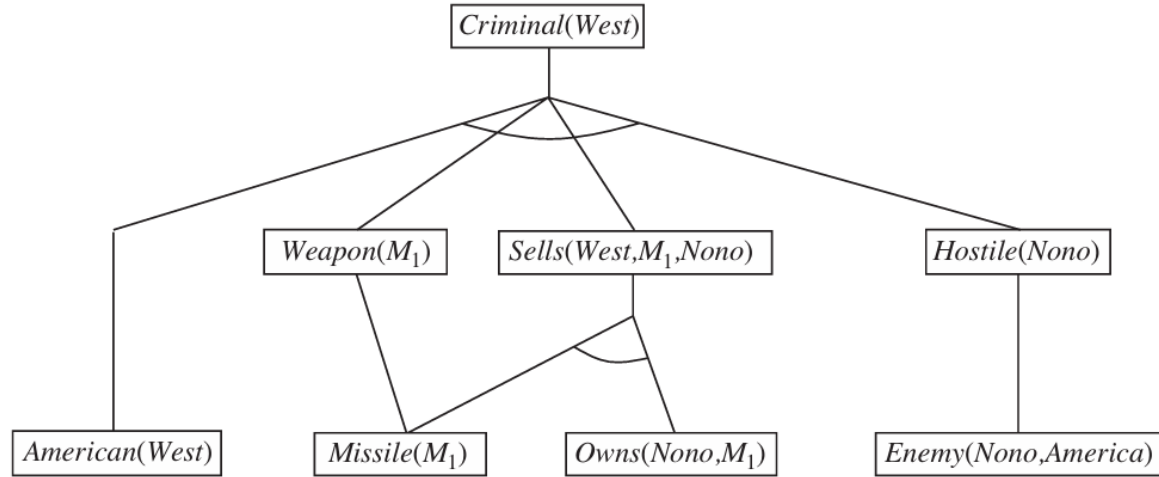
$American(West)$

$Enemy(Nono, America)$

Forward Chaining ASK

Iteration 1:

- R5 satisfied with $\{x/ M_1\}$ and
R9: Sells(West, M_1 , Nono) is added
- R4 satisfied with $\{x/ M_1\}$ and
R10: Weapon(M_1) is added
- R6 satisfied with $\{x/\text{Nono}\}$ and
R11: Hostile(Nono) is added



Iteration 2:

- R1 is satisfied with $\{x/\text{West}, y/ M_1, z/\text{Nono}\}$ and **Criminal(West) is added.**

Conclusion

- Once we have facts that evaluate to T or F
- We can apply Forward Chaining, Backwards Chaining and
- Resolution
- The key is to understand Unification
- Very similar to Logical agents.

Thank You

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